REPUBLIQUE DU CAMEROUN

PAIX-TRAVAIL-PATRIE

MINISTRE DE L'ENSEIGNEMENT SUPERIEUR

FACULTE DE L'INGINEURIE ET DE TECHNOLOGIES

REPUBLIC OF CAMEROON

PEACE- WORK -FATHERLAND

MINISTER OF HIGHER EDUCATION

FACULTY OF ENGINEERING AND TECHNOLOGY

UNIVERSITY OF BUEA

COURSE CODE : CEF440

COURSE TITLE : INNTERNET PROGRAMING AND MOBILE DEVELOPMENT

Date: 19-05-2024

DEPARTMENT OF COMPUTER ENGINEERING
TASK 4: SYSTEM MODELLING AND DESIGN

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1. INTRODUCTION

A. FOREWORD

In todays fast pace technological landscape, businesses and organizations rely on robust systems to streamline their operations and enhance productivity. The aim is for innovation to meet efficiency, the simple answer to this? system modelling and design.

System modelling and design is the process of creating a blueprint that captures the structure, behavior and functionality of a system, allowing analysis, simulation and optimization of complex systems to ensure they meet the desired requirements and objectives.

B. RELATION TO SCOPE

In relation to the fingerprint-based attendance recording system, this cutting-edge solution leverages biometric technology to accurately record and track the attendance of individuals by meticulously expatiating on architectural designs and workflows as well as ensuring seamless integration with existing infrastructure.

The boons are thus optimization of recording process, elimination of manual errors and overall enhancement in security just to name a few.

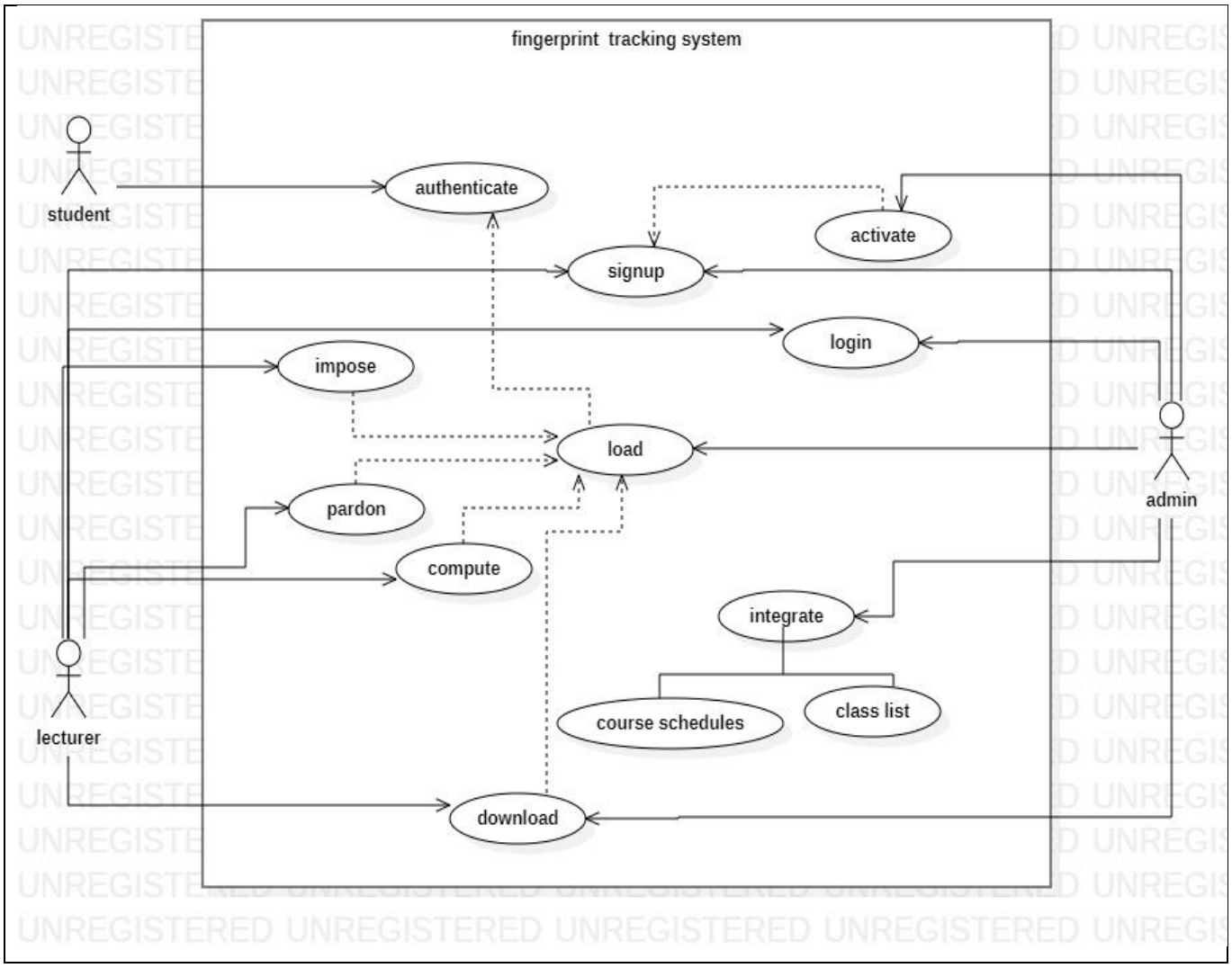
This report covers system modelling and design aspects such as

- Use Case Analysis and Design
- Class Diagram and Design
- Sequence Diagram and Design
- Context Diagram and Design
- Deployment Diagram and Design

2. USE CASE ANALYSIS AND DESIGN

A Use Case explains how users interact with a digital product of system by outlining the flow of user inputs, establishing successful and failed paths to meeting goals thus allowing better understanding of why errors occur. The result is a better understanding of what the system does and how it performs said tasks.

Attached below is a Use Case Diagram showing various user role participants and their interaction with system.



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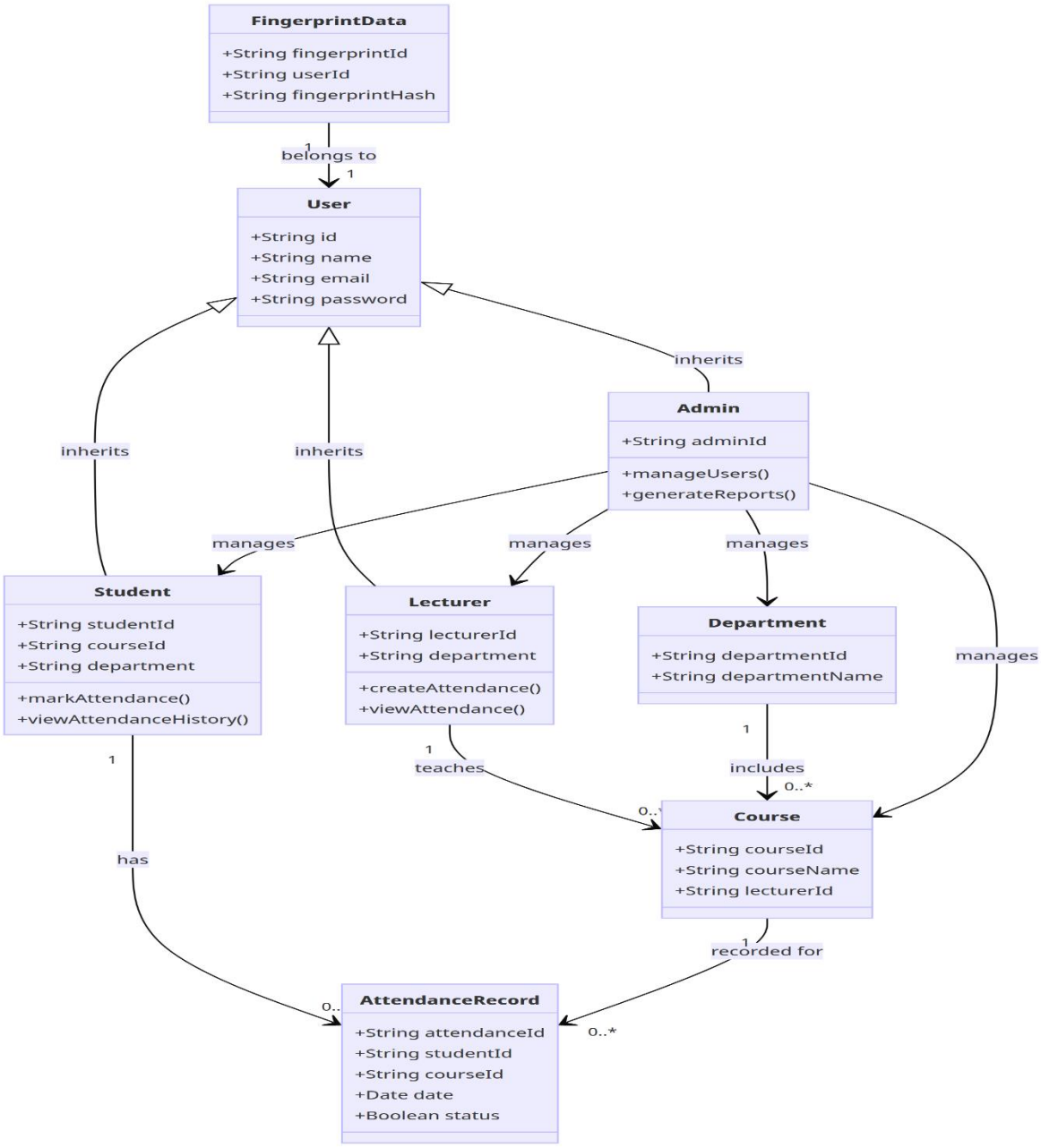
3. CLASS DIAGRAM AND DESIGN

Class diagrams model the objects that make up the system, displaying inter-object relationship and what said objects do and provide as services. Thus, this UML diagram provides the blueprint for the system under study (biometric/fingerprint attendance tracking system)

A. DIAGRAM



B. DESCRIPTION

a) **Classes:** This species the various classes

- User: A base class representing common attributes for all users (students, lecturers, and admins).
- Student: Inherits from User and adds specific attributes and methods for students.

- Lecturer: Inherits from User and adds specific attributes and methods for lecturers.
- Admin: Inherits from User and adds specific attributes and methods for admins.
- FingerprintData: Stores fingerprint information associated with users.
- AttendanceRecord: Records attendance data.
- Course: Represents a course with relevant attributes.
- Department: Represents a department with relevant attributes.

b) Relationships

I. Inheritance

- Student, Lecturer, and Admin inherit from User.

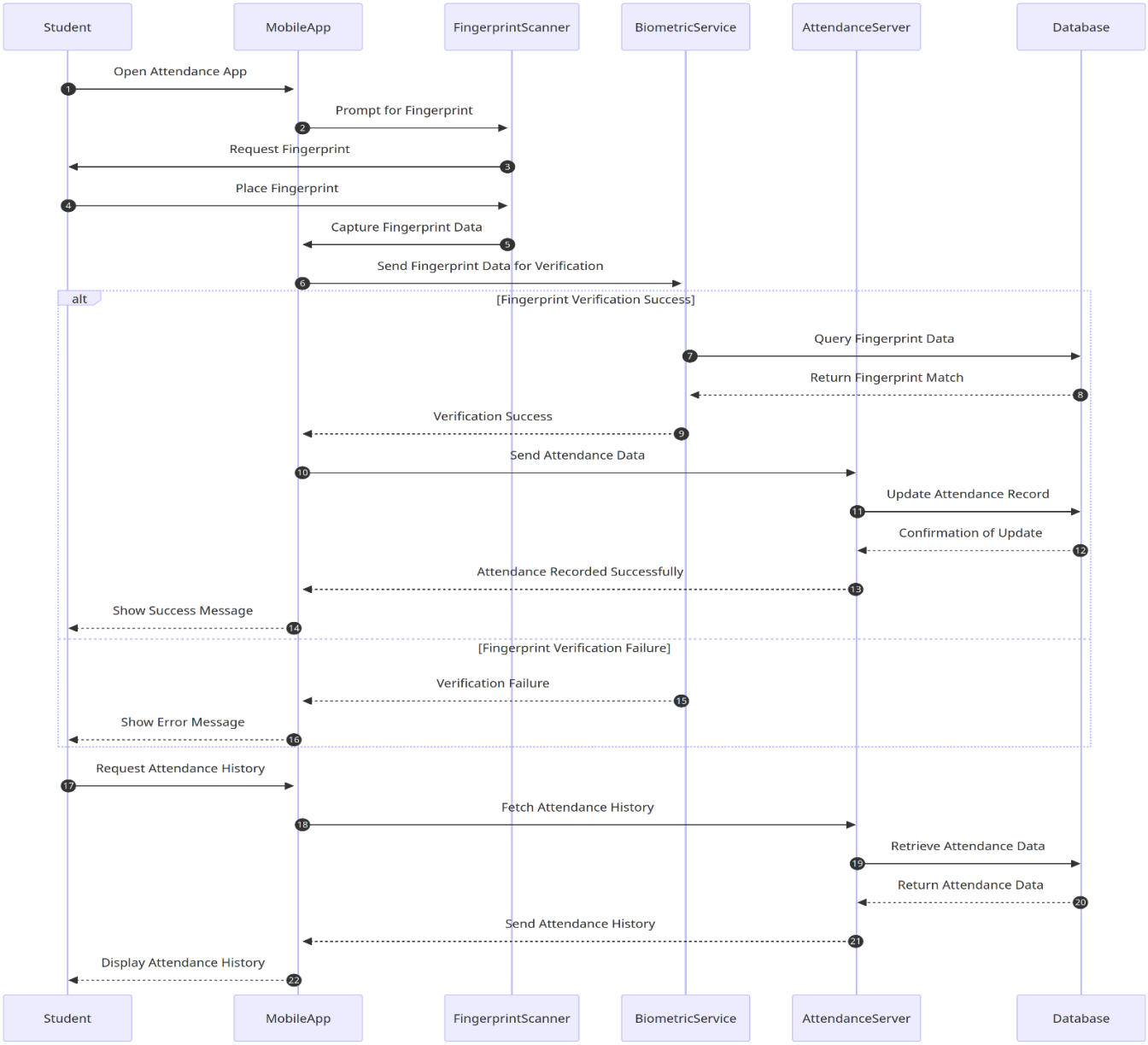
II. Associations

- A Student can have multiple AttendanceRecord entries (1-to-many).
- A Lecturer can teach multiple Course instances (1-to-many).
- A Department can include multiple Course instances (1-to-many).
- A Course can have multiple AttendanceRecord entries (1-to-many).
- Each FingerprintData entry is associated with one User (1-to-1).
- An Admin manages multiple entities such as Student, Lecturer, Course, and Department

4. SEQUENCE DIAGRAM

This diagram specifies the interaction between objects as well as message flows across the life line. The sequence diagram of our target system outlines the process flow for a biometric (fingerprint) school attendance mobile app detailing the interactions between the student, mobile app, and backend services for checking attendance as well as retrieving attendance history.

A. Diagram



B. BREAK DOWN OF THE DIAGRAM

- Opening the App

The lecturer opens the attendance app on their mobile device.

- Fingerprint Scanning

The mobile app prompts the student to provide their fingerprint.

The fingerprint scanner requests the fingerprint from the student, captures it, and sends the data back to the mobile app.

- Fingerprint Verification

The mobile app sends the captured fingerprint data to the biometric service for verification.

The biometric service queries the database to match the fingerprint data.

- Handling Verification Result

i. Success

If the fingerprint matches, the biometric service returns a success message to the mobile app.

The mobile app sends the attendance data to the attendance server.

The attendance server updates the attendance record in the database.

The database confirms the update, and the success message is propagated back to the student.

ii. Failure

If the fingerprint verification fails, the biometric service returns a failure message to the mobile app, which then shows an error message to the student.

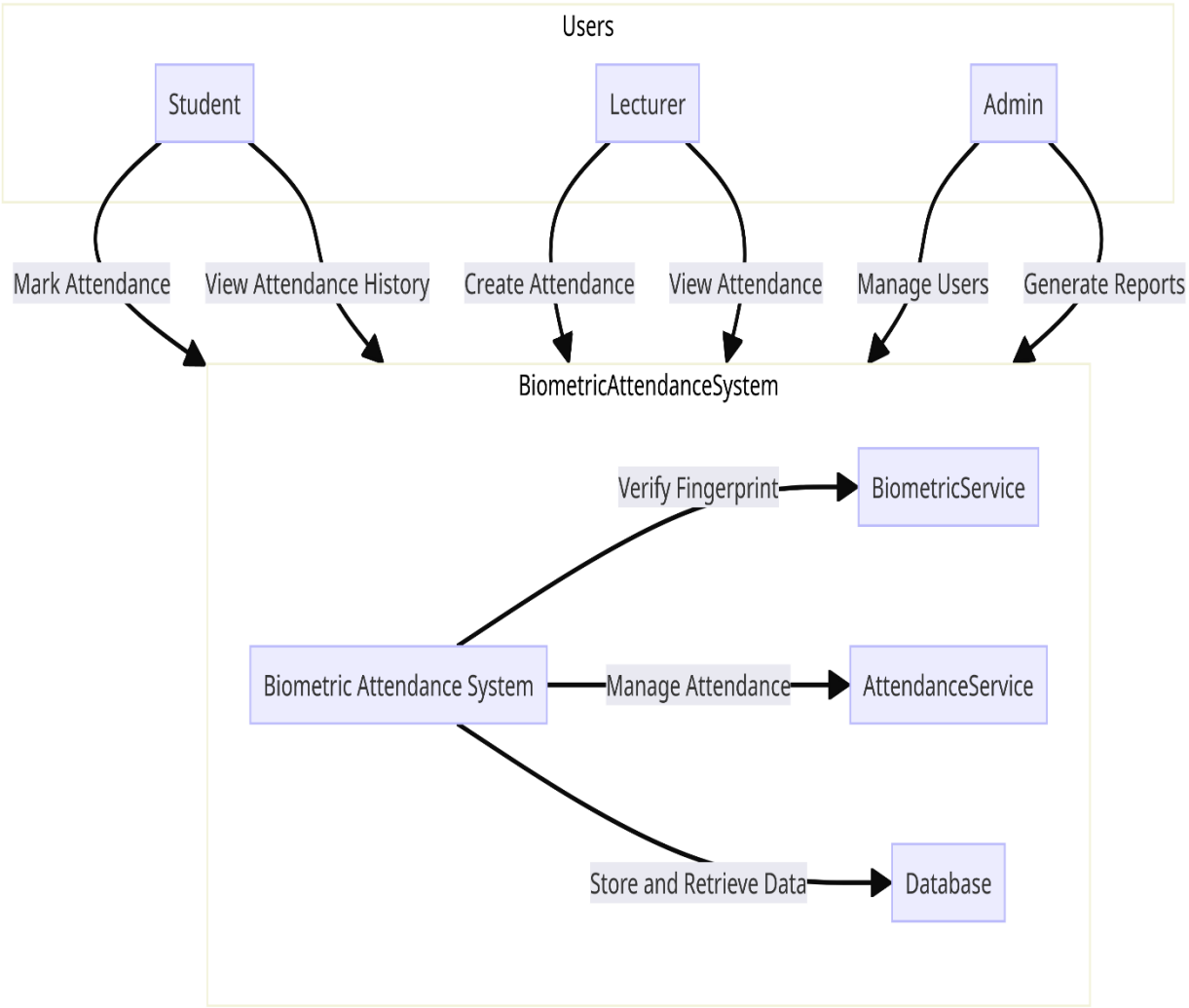
- Fetching Attendance History

I. The student can request to view their attendance history.
II. The mobile app requests this data from the attendance server.
III. The attendance server retrieves the data from the database and sends it back to the mobile app for display.

5. CONTEXT DIAGRAM AND DESIGN

A context diagram, also known as a context-level data flow diagram (DFD), is a visual representation that illustrates the interactions between a system or software application and its external entities by providing a high-level overview of the system's boundaries; showing how it interacts with external entities, processes, and data flows. Here, the aim is to depict the attendance system's scope and context, helping to understand the former's interactions with its environment.

A. DIAGRAM



B. DESCRIPTION

- **Components**

I. Student, Admin, and Lecturer: The three types of users interacting with the system.
II. Biometric Attendance System: The main system where all interactions occur.
III. Biometric Service: A subsystem within the Biometric Attendance System responsible for verifying fingerprints.
IV. Attendance Service: A subsystem responsible for managing attendance records.
V. Database: The storage system for all data, including user information, fingerprint data, and attendance records.

- **User Interactions**

  - Student: Marks attendance by interacting with the Biometric Attendance System.
  - Views attendance history through the Biometric Attendance System.
  - Lecturer: Creates attendance records within the Biometric Attendance System. Views attendance records.
  - Admin: Manages users, including students and lecturers, through the Biometric Attendance System. Generates reports related to attendance and user activities.

- **System Interactions**

The Biometric Attendance System interacts with;

a) Biometric Service to verify fingerprints.
b) Attendance Service to manage attendance records.
c) Database to store and retrieve user data, fingerprint data, and attendance records.

C. PURPOSE OF CONTEXT DIAGRAM

I. **High-Level Overview**: Provides a clear, simplified view of the system's architecture and the interaction between different components.

II. **Communication Tool:** Useful for communicating the system design to stakeholders, including developers, clients, and project managers.

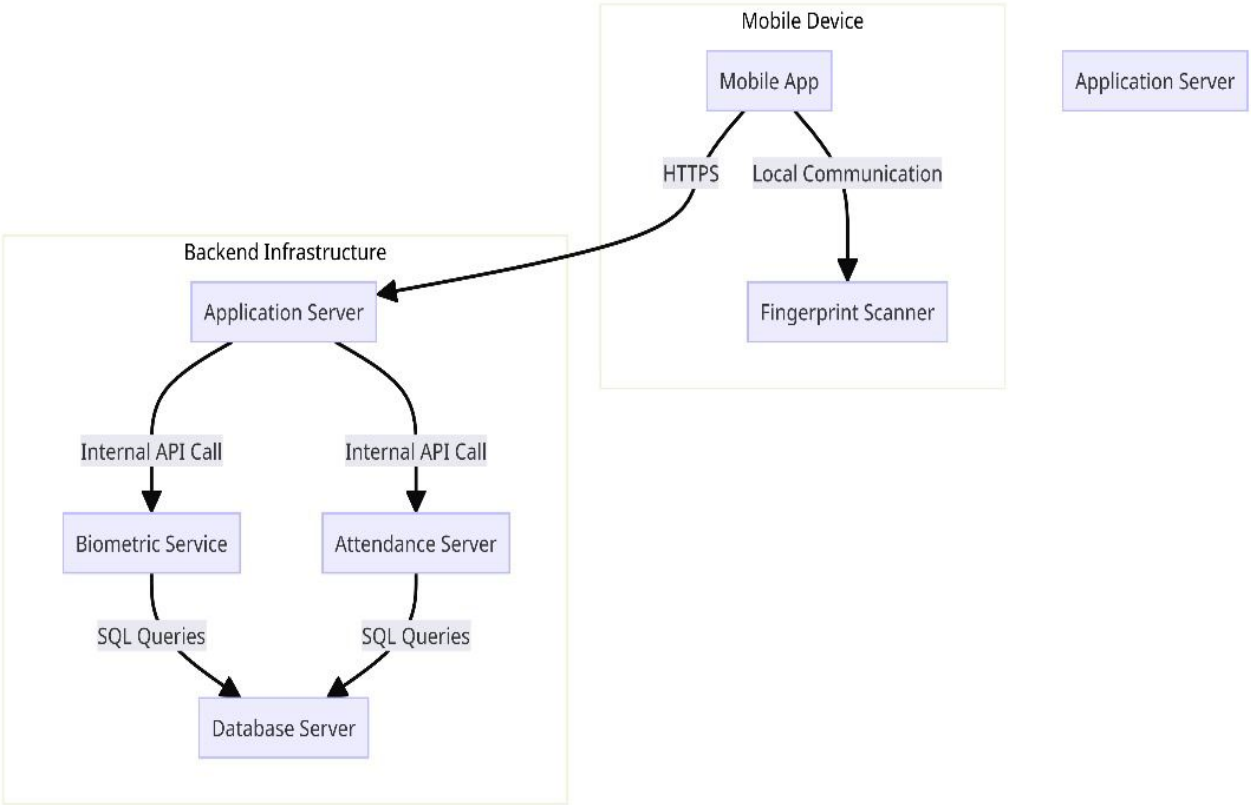
III. **Foundation for Detailed Design**: Acts as a foundation for creating more detailed diagrams and design documents, such as sequence diagrams, class diagrams, and deployment diagrams.

This context diagram thus helps in understanding the overall structure and data flow of the biometric (fingerprint) school attendance mobile app, ensuring that all stakeholders have a shared understanding of the system.

6. DEPLOYMENT DIAGRAM AND DESIGN

The deployment diagram is used to visualize the topology of the physical components of the system where the software components are deployed. The deployment diagram of this system will look as follows;

A. DIAGRAM



B. EXPLANATION

- **Components**

  I. **Biometric device:** This is the user's smartphone or tablet running the biometric student attendance app.

  II. **Fingerprint Sensor:** Integrated into the mobile device, this sensor captures and scans the user's fingerprint.

  III. **Database:**

  This component stores all student-related data, including fingerprints, names, and enrollment numbers.

  IV. **Server:** Acts as a centralized hub for data synchronization and communication between the mobile app and database. Servers here include:

  a) Application Server: is the server involved with hosting both the Biometric Service and the Attendance Server.

  b) Biometric Server: meanwhile is responsible for verifying the fingerprint data.

  c) Attendance Server: is the service tasked with recording and retrieving attendance data

C. BENEFITS

The deployment diagram of this biometric student attendance mobile app offers several benefits:

- Improved Accuracy:

  Fingerprint scanning ensures precise identification of students.

- Enhanced Security:

  It eliminates unauthorized access to sensitive information.

- Time Efficiency:

  Attendance tracking becomes faster and more convenient for both students and administrators.

- Data Privacy:

  Student data remains secure as it is encrypted during transmission.

By utilizing a deployment diagram to illustrate its components and interactions, users can better understand how this system works to improve accuracy, security, efficiency, and privacy in

7. CONCLUSION

To conclude this, we observe that correct and detailed modelling of the system greatly improves development as the main picture becomes clearer. Modularity is also an advantage of system modelling and design due to splitting of the system to understandable subs and hence, exploiting the functional, non-functional and technical requirements observed in the requirement gathering and analysis phases.